

DIEGO BERMUDEZ

412-519-6530 ◊ dabermud@andrew.cmu.edu ◊ linkedin.com/in/dabermudezs

SUMMARY

- AI Scientist with experience building and deploying ML models across large-scale financial and mobility datasets, combining classical ML, NLP, and deep learning for measurable impact.
- Strong in end-to-end applied ML: data discovery, ETL, feature engineering, experimentation (A/B testing), model evaluation, and productionization with reproducible pipelines.
- Hands-on with Python, SQL, Linux, cloud data platforms, and modern ML tooling to deliver personalization, risk, and customer analytics solutions.

EDUCATION

Carnegie Mellon University, Heinz College, Pittsburgh, PA **December 2025**
Master of Information Systems Management – Business Intelligence and Data Analytics

- **GPA:** 3.94/4.00
- **Relevant courses:** AI Engineering, Deep Learning, Applications of LLMs, Distributed Systems.

University of Los Andes, Bogotá, Colombia **December 2022**
Master of Arts in Economics

- **Thesis:** Access to financial services as a mechanism for consumption smoothing in the context of violence.
- **Relevant courses:** Advanced Econometrics, Topics in Machine Learning, Financial Risk Analysis.

University of Los Andes, Bogotá, Colombia **July 2020**
Bachelor of Arts in Economics

- **Relevant courses:** Data Mining and Applications, Financial Microeconometrics, Advanced Macroeconomics.

SKILLS

- **Programming Languages:** Python, SQL, R, Java, VBA, STATA, MATLAB.
- **Machine Learning & AI:** scikit-learn, PyTorch, TensorFlow, XGBoost, Transformers, neural networks, recommendation systems, anomaly detection, clustering, classification, regression, time series, text mining, NLP.
- **Data Engineering & Large-Scale Data:** ETL pipelines, feature engineering, Spark, SQL optimization, PostgreSQL, BigQuery, Hadoop concepts, data validation, data quality checks.
- **MLOps & Reproducibility:** MLflow, Docker, CI/CD pipelines, model packaging, experiment tracking, monitoring-ready artifacts.
- **Cloud & Systems:** GCP, AWS (exposure), Linux, Git, REST APIs.
- **Visualization & BI:** Power BI, Tableau, Looker Studio, GIS.

PROFESSIONAL EXPERIENCE

Davivienda Bank (Daviplata) **Bogotá, Colombia**
Data Scientist **April 2023 – August 2024**

- Built and deployed geospatial ML pipelines to infer SMB locations from transactional and operational signals, increasing geographic coverage from **86% to 90%** and improving acquisition from **8.75% to 10.5%** with **86% precision** (**Python, scikit-learn, GIS, Docker, MLflow, PostgreSQL**).
- Led feature engineering and dataset assembly across cloud sources to enable segmentation and targeting at scale, supporting SMB base growth from **1.6M to 2.5M**, **97% YoY** activity growth, and billing increase from **COP 214MM to COP 549MM** (**BigQuery, GCP, SQL**).
- Designed and analyzed A/B experiments (2 to 3 weeks; **10K to 150K** customers) and quantified causal lift on customer outcomes, improving engagement and savings transactions from **8% to 12%** (**Python, SQL, experimentation**).
- Contributed to a recommendation and offer optimization framework for credit and wallet products, improving conversion by **19%** via lifecycle-aware features and policy-constrained ranking logic (**scikit-learn, PySpark, GCP**).
- Partnered with Data Engineering to productionize model-ready datasets with standardized schemas, automated ingestion, and QA checks to improve reliability and reuse across analytics and modeling (**ETL, testing, BigQuery**).

Davivienda Bank **Bogotá, Colombia**
Business Intelligence Analyst **November 2021 – April 2023**

- Built automated data pipelines and validation logic for **50K+** monthly regulatory petitions, reducing errors to **1K per month**, saving **10 hours per week**, and strengthening auditability through traceable reporting (**Python, SQL, ETL**).
- Reduced audit findings by **75%** by improving reconciliation, KPI logic, and governance controls across reporting workflows.
- Developed an NLP quality model for internal operations, processing **35K chats/day** with **88%+ accuracy** and extracting **20+** features for performance monitoring across **8 teams** (**spaCy, scikit-learn, SQL, Python**).
- Delivered stakeholder-facing dashboards with historical tracking and performance optimization to support operational decision-making and continuous improvement (**Power BI, Looker Studio**).

The Barometer **Bogotá, Colombia**
Data Analyst **December 2020 – January 2022**

- Built NLP and text mining pipelines over **1M** posts and articles per year, applying topic modeling and clustering to identify emerging themes and narrative shifts (**Python, TF-IDF, LDA, clustering**).
- Developed network-based analytics to map diffusion and influence patterns, translating signals into actionable insights for multi-country stakeholders.
- Delivered monthly bulletins and 8 policy briefs across **9 countries**, turning large-scale unstructured data into decision-ready intelligence.

Ecopetrol S.A. – Business & Commercial VP **Bogotá, Colombia**
Crude Trading Intern **January 2020 – July 2020**

- Developed KPI models and dashboards integrating CRM and ERP sources, improving visibility and reducing manual reporting time by **25%** (**SQL, Excel, Python, Power BI**).

ACADEMIC EXPERIENCE

Carnegie Mellon University - Mobility Analytics Center
Research Assistant

Pittsburgh, PA
November 2024 – Present

- Built large-scale ETL and geospatial feature engineering pipelines to study work zone impact on crash risk, integrating **80K+** work zones, **500K** road segments, and **1K to 3K** crashes (**Python, GeoPandas, PostgreSQL/PostGIS, Linux**).
- Processed **3 TB** of multi-source data and reduced processing time by **60%** through optimized joins, batching strategies, and efficient spatial matching (**SQL, Python**).
- Applied supervised learning and causal ML methods for heterogeneous effects analysis, producing model-ready datasets and research-grade evaluation artifacts (**scikit-learn, TensorFlow, econml, R**).

Carnegie Mellon University – FLM Haiti
Capstone Project

Pittsburgh, PA
August 2025 – December 2025

- Designed a hybrid deployment strategy for an offline-first EMR system, aligning data workflows for low-connectivity settings and defining production readiness steps (**Flutter, Supabase, PostgreSQL**).
- Integrated OCR-assisted digitization to extract structured patient data with **70%** average accuracy, supporting **400** records per month for a patient population of **30K** (**Azure Forms OCR, data pipelines**).

University of Los Andes
Teaching Assistant – Financial Regulation

Bogotá, Colombia
January 2022 – July 2022

- Mentored students in applied financial regulation projects, guiding data-backed recommendations under compliance and policy constraints.

Center for Economic Development Studies (CEDE), University of Los Andes
Professional Research Assistant

Bogotá, Colombia
November 2020 – July 2021

- Applied structural break methods and clustering techniques to quantify mobility shifts during COVID-19, combining econometric modeling with ML-based segmentation (**R, Python, STATA**).
- Built automated data collection and transformation pipelines to construct external indicators for modeling and hypothesis testing (**Python, BeautifulSoup, Pandas, NumPy**).

AWARDS

- Merit-Based Scholarship – Heinz College, Carnegie Mellon University (Mar 2024)
- Asobancaria 7th Call for Papers – Best Paper in Economics (Sep 2023)
- COLFUTURO Loan-Scholarship – Academic Merit and Professional Excellence (May 2023)